When Does Collateralized Borrowing Become Contractionary? – An Endogenous Debt Threshold

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Abstract

Housing collateral enables households to borrow, whereas accumulated debt discourages demand due to subsequent debt service absorbed by borrowers' consumption rather than their saving. Existing models capture this duality but do not identify a critical debt level at which the marginal effect of borrowing changes sign. This paper shows that, given the non-monotonic marginal expenditure effect, strict concavity of net expenditure in the debt stock delivers a unique expansionary–contractionary threshold $\bar{D}(\theta)$, which is endogenously generated by a risk-premium specification. In addition, this paper finds that a larger creditor income share θ or more unequal economy lowers the threshold, and a binding collateral constraint maps it into a critical collateral price $\bar{q}(\theta)$. Under a collateral-price closure, $\bar{D}(\theta)$ is where a credit boom ceases to accelerate.

Keywords: household debt, collateralized borrowing, effective demand, income distribution, debt threshold, collateral constraint, credit cycles, financial instability
JEL classification: E21, E32, E44, D31, O41

1. Introduction

Household borrowing has opposing effects on aggregate demand. New credit relaxes liquidity constraints and encourages current expenditure; however, the resulting debt stock shrinks future demand as debt service obligation

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weigh on borrowing households. Collateralized borrowing reinforces the first mechanism, since rising asset values expand borrowing capacity (Kiyotaki and Moore, 1997; Iacoviello, 2005).[1]

The second mechanism is not an accounting necessity. It depends on how debtor households absorb their servicing obligations. Under the "pecking order" specification of Setterfield and Kim (2016), in which workers protect consumption and reduce saving, which they consider a luxury, first, servicing transfers income that would otherwise have leaked from the circular flow, and the marginal effect of the debt stock on demand is still positive rather than negative. This paper assumes the conventional case, in which servicing crowds out borrowers' consumption. Even under that assumption, the paper shows that the question of primary interest is not whether the net effect of debt is expansionary or contractionary, but rather the debt level at which the marginal effect of additional borrowing reverses sign.

This duality has long been recognized. Palley (1994) combines the initial effect of demand expansion by household borrowing with the subsequent contractionary effect of debt service payment on aggregate demand, in light of Minsky (1977, 1982), for which prolonged economic stability breeds the optimism that leads investors and bankers to take on excessive debt. Dutt (2006) demonstrates that short-term boost to demand and output from household borrowing may be temporary as accumulated debt weakens it through interest transfers and income redistribution, using a Steindlian framework (Steindl, 1976[1952], 1979). In a New Keynesian setting, Eggertsson and Krugman (2012) similarly show how excessive debt and forced deleveraging can depress aggregate demand and generate a liquidity trap. These dynamics directly reflects the Post-Keynesian notion of increasing financial commitments and risk (Kalecki, 1937; Mott, 2009) and relates to indebted-demand analysis, which emphasizes transfers from higher-spending borrowers to lower-spending creditors (Mian et al., 2021).

In these specifications, however, the marginal demand effect of the debt stock does not endogenously change its sign while the level of indebtedness rises. In Palley (1994, p. 381), for example, the flow and stock effects of debt enter through the coefficients $b_2 > 0$ and $b_3 r < 0$ of a second-order difference equation. This suggests that marginal effect associated with the debt stock

[1] Kiyotaki and Moore (1997) concerns firms borrowing against landholdings used in production rather than households.

should be fixed by the model's parameters. In Dutt (2006, pp. 348–355), the short-run utilization rate is affine in the debt–capital ratio δ, implying a signed and constant marginal effect $u_\delta < 0$. Higher desired borrowing is expansionary in the short run, whereas its long-run effect on accumulation hinges on the parameter condition $g \gtreqless si$. This is a parameter-dependent comparative-static condition rather than a critical level of the debt stock at which its marginal demand effect changes sign.

Related stock–flow consistent models generate substantially richer dynamic interactions, but none of them characterizes an endogenous debt level at which the marginal demand effect of indebtedness changes sign. In Ryoo and Kim (2014), the effects of the debt ratio on the profit share and on workers' consumption are ambiguous in sign but, in the reported comparative statics, do not exhibit a debt-level-dependent sign reversal, and the goods-market closure absorbs changes in indebtedness through distribution rather than through the level of demand.[2] In Ryoo (2016), the marginal effect of debt on aggregate consumption is likewise constant in the relevant comparative static, while the collateral effect remains positive throughout the admissible range of housing wealth. In Setterfield and Kim (2016), debt servicing raises short-run utilization and growth under the saving-first specification.

Nonlinearity is therefore not absent from this literature; it enters through a different object. Bounded emulation functions in Ryoo and Kim (2014) and bounded desired-portfolio functions in Ryoo (2016) introduce curvature in other parts of the dynamic system in order to bound trajectories and generate closed orbits. Setterfield and Kim (2016) do obtain an interior turning point in the debt ratio, but it occurs in the law of motion of debt rather than in expenditure: their $\dot{d}_R$ schedule is inverse-U shaped because accumulation is itself increasing in d_R. The relevant distinction is therefore not between linear and nonlinear models, but between nonlinearity elsewhere in the dynamic system and a debt-level-dependent reversal in the marginal debt–demand relation itself. None of the models reviewed here delivers the latter; the present paper does, deriving a unique endogenous debt threshold $\bar{D}(\theta)$ at which the marginal demand effect of additional borrowing changes

[2] Their footnote 27, p. 598, notes that in Dutt (2006, 2008) and Isaac and Kim (2013) the short-run effect of higher household indebtedness is unambiguously contractionary, owing to a different specification of consumption behaviour. Unambiguous or not, the sign in those models is fixed rather than dependent on the level of the debt stock.

sign. Table 1 summarizes the comparison.

This distinction is empirically relevant. Lombardi et al. (2017) find that household debt supports consumption and output growth in the short run but has increasingly adverse long-run effects at high debt-to-GDP ratios.[3] Related historical evidence associates more credit-intensive expansions with deeper recessions and slower recoveries (Jordà et al., 2013). Such evidence motivates a debt effect whose marginal contribution varies with the level of indebtedness.

This paper identifies the nonlinear condition required to generate that object. Given an initially positive and eventually negative marginal expenditure effect, strict concavity of the net expenditure function in the debt stock makes the marginal effect strictly decreasing and therefore delivers a unique threshold $\bar{D}(\theta)$. A simple risk-premium specification derives this concavity even when the gross marginal expenditure benefit of credit is constant; the concavity is thus a consequence of the lending technology rather than an assumption imposed on expenditure directly. The analysis further shows that a larger creditor income share lowers $\bar{D}(\theta)$ when additional creditor income accrues to progressively lower-spending households. Finally, when the collateral constraint binds, the debt threshold maps into a critical collateral price $\bar{q}(\theta)$, so that the demand effect of collateral appreciation itself changes sign at the threshold. This last result is the sharpest point of departure from Ryoo (2016), the closest antecedent, in which collateral appreciation raises borrowing, consumption and the profit share at every level of housing wealth.

The threshold also has a dynamic reading. Embedding it in a collateral-price closure shows that $\bar{D}(\theta)$ marks the inflection of a credit boom rather than its end: along an expanding trajectory, debt continues to accumulate beyond the threshold at a decelerating rate until the upper stationary point is reached, while a larger creditor income share compresses the accelerating phase from both ends.

[3] Their estimates indicate stronger adverse long-run effects above household debt ratios of approximately 60 percent of GDP for consumption and 80 percent for output growth. These estimates motivate, rather than identify, the endogenous threshold derived here.

Table 1: Marginal demand effect of the debt stock and the thresholds delivered

Study	Relevant marginal effect of indebtedness	Behaviour in the debt stock	Threshold delivered
Palley (1994)	$\partial y_t/\partial \Delta D_t = b_2 = 1 - a_2 > 0$; $\partial y_t/\partial D_{t-1} = b_3 r < 0$, with $b_3 = a_2 - a_1$ (eq. 7)	constant	maximum leverage ratio a_3; dynamic stability boundary (eq. 11.2′)
Dutt (2006)	$u = [g - (s+\beta)i\delta]/\Gamma$ (eq. 10), so $u_\delta = -(s+\beta)i/\Gamma < 0$; also $u_\beta > 0$ (p. 349)	constant	parameter condition $g \gtreqless si$ governing the long-run effect of a change in desired borrowing
Ryoo and Kim (2014)	$\pi_m = \dfrac{f_1 r + f_2(1+\alpha) + n - r(1+\phi_1)}{u\Delta} \gtreqless 0$; $c_m^w < 0$ (eqs. 27, 29)	constant[a]	$\lvert\psi_1\rvert\phi_2$ condition; local instability and limit cycle
Ryoo (2016)	Debt: $\chi_m = [r f_y + f_\omega(1+\alpha)] + [n - r(1+\mu_y) - \mu_\omega] \gtreqless 0$; collateral: $\chi_{h^w} = \mu_\omega > 0$ (eq. 25)	constant	Hopf bifurcation in the expectation-adjustment speed ν
Setterfield and Kim (2016)	$\partial u/\partial d_R > 0$, with u affine in d_R (eq. 21)	constant, positive	debt-service feasibility ceiling $d_R^{\max}$, not a demand sign-switch threshold
This paper	$F_D(D,\theta)$ with $F_{DD}(D,\theta) < 0$ (Assumption 2)	strictly decreasing	unique demand sign-switch threshold $\bar{D}(\theta)$; critical collateral price $\bar{q}(\theta)$ under a binding constraint

Note: The models differ in closure, so the relevant demand-side object differs across rows—output in Palley (1994), utilization in Dutt (2006) and Setterfield and Kim (2016), and distribution and consumption in Ryoo and Kim (2014) and Ryoo (2016). The entries report the debt-stock marginal effect relevant to each model; the Palley (1994) and Ryoo (2016) rows additionally report, respectively, the borrowing-flow and collateral effects for comparison.

[a] The rentier consumption function is homogeneous of degree one, so f_1, f_2 are formally functions of y^r/ω^r and hence of m. This channel is not exploited: the reported comparative statics and the simulations treat f_1, f_2 as constants.

2. Model

2.1. Households, Collateral, and Aggregate Demand

Consider an economy with a borrower sector and a heterogeneous creditor sector. Let $\theta \in (0,1)$ denote the share of aggregate income accruing to creditors. Borrowers have marginal propensity to consume c_b, while $c_c(\theta)$ denotes the average marginal propensity to consume among creditors. Following Kaldor (1955), we assume

$$0 < c_c(\theta) < c_b < 1, \qquad c_c'(\theta) < 0. \tag{1}$$

The condition $c_c'(\theta) < 0$ captures a compositional effect: as the creditor income share rises, additional creditor income accrues to progressively lower-spending households. The aggregate marginal propensity to consume is

$$c(\theta) = (1-\theta)c_b + \theta c_c(\theta), \tag{2}$$

so that

$$c'(\theta) = c_c(\theta) - c_b + \theta c_c'(\theta) < 0. \tag{3}$$

Since $0 < c(\theta) < 1$, the Keynesian multiplier

$$\mu(\theta) = \frac{1}{1 - c(\theta)} \tag{4}$$

is positive and satisfies $\mu'(\theta) < 0$. A larger creditor income share therefore reduces the aggregate expenditure multiplier, reflecting Kaldor (1955)'s seminal finding.

Household borrowing is collateral constrained. Similarly to Iacoviello (2005), suppose the constraint binds:

$$D = mqH, \tag{5}$$

where D is household debt, q the collateral price, H the collateral stock, and $m \in (0,1)$ the loan-to-value ratio.

Aggregate demand is

$$Y = \mu(\theta)\left[A + F(D,\theta)\right], \tag{6}$$

where A denotes autonomous expenditure and $F(D, \theta)$ is the net expenditure effect of household debt.[4] The function F captures the expenditure enabled by borrowing net of debt servicing and associated balance-sheet adjustment that is not recycled into current demand. Throughout, F measures the first-round expenditure effect of household debt, which the multiplier $\mu(\theta)$ subsequently amplifies. In this static framework, both the expenditure and servicing effects are expressed in terms of the debt level.

Assume that $F : \mathbb{R}_+ \times (0, 1) \to \mathbb{R}$ is twice continuously differentiable. The threshold result requires the following conditions.

Assumption 1 (Initial expansion)**.**

$$F_D(0, \theta) > 0. \tag{7}$$

At low indebtedness, additional borrowing raises expenditure.

Assumption 2 (Declining marginal expenditure effect)**.**

$$F_{DD}(D, \theta) < 0. \tag{8}$$

The marginal expenditure effect of borrowing declines strictly with the debt stock.

Assumption 3 (Distributional leakage)**.**

$$F_{D\theta}(D, \theta) < 0. \tag{9}$$

A larger creditor income share reduces the marginal expenditure effect of additional borrowing.

Assumption 4 (Eventual contraction)**.**

$$\lim_{D \to \infty} F_D(D, \theta) < 0. \tag{10}$$

At sufficiently high indebtedness, the marginal financial burden of additional borrowing dominates its expenditure effect.

[4] See Appendix A for its derivation.

2.2. A Microfoundation for Concavity

The nonlinear restriction in Assumption 2 can arise from an endogenous servicing burden rather than from diminishing gross expenditure benefits. Consider

$$F(D, \theta) = B(D) - \lambda(\theta) r(D) D, \tag{11}$$

where $B(D)$ denotes expenditure enabled by borrowing, $\lambda(\theta)$ the fraction of debt-service income not recycled into current expenditure, and $r(D)$ the effective borrowing rate.

In the baseline regime, debt service is met out of borrowers' consumption, so that servicing transfers income from a marginal propensity c_b to the lower propensity $c_c(\theta)$ and

$$\lambda(\theta) = c_b - c_c(\theta) > 0, \qquad \lambda'(\theta) = -c_c'(\theta) > 0. \tag{12}$$

Hence the same compositional change that lowers the aggregate multiplier in Equation (4) also raises the expenditure leakage associated with debt service. The alternative regime, in which servicing is met out of saving, is considered below.

The effective borrowing rate is decomposed as $r(D) = r_f + \rho(D) > 0$, where r_f is a benchmark funding rate, taken as given, and $\rho(D)$ is a leverage-sensitive credit risk premium. The specification does not require the observed mortgage rate to rise with aggregate household debt. It requires only that, holding benchmark funding conditions fixed, the credit risk premium associated with additional leverage rise with indebtedness.

Suppose

$$B'(D) > 0, \qquad B''(D) \leq 0, \qquad \rho'(D) > 0, \qquad \rho''(D) \geq 0. \tag{13}$$

Differentiating Equation (11) and using $r(D) = r_f + \rho(D)$ yields

$$F_D(D, \theta) = B'(D) - \lambda(\theta) \left[r_f + \rho(D) + D\rho'(D)\right], \tag{14}$$

$$F_{DD}(D, \theta) = B''(D) - \lambda(\theta) \left[2\rho'(D) + D\rho''(D)\right] < 0, \tag{15}$$

$$F_{D\theta}(D, \theta) = -\lambda'(\theta) \left[r_f + \rho(D) + D\rho'(D)\right] < 0. \tag{16}$$

Assumptions 2 and 3 follow directly, and Assumption 1 holds throughout the admissible distributional domain whenever

$$B'(0) > \sup_{\theta \in (0,1)} \lambda(\theta) \left[r_f + \rho(0)\right]. \tag{17}$$

This condition guarantees an initially positive marginal expenditure effect for every admissible θ.

Assumption 4 also follows from Equation (13). Since $\rho''(D) \geq 0$, the premium schedule $\rho'(D)$ is nondecreasing and bounded below by $\rho'(0) > 0$, so that $D\rho'(D) \geq D\rho'(0)$ and

$$r_f + \rho(D) + D\rho'(D) \longrightarrow \infty. \tag{18}$$

Meanwhile, $B''(D) \leq 0$ implies that $B'(D)$ is bounded above by $B'(0)$. Consequently,

$$F_D(D, \theta) \longrightarrow -\infty. \tag{19}$$

The specification therefore generates a strictly declining and eventually negative marginal expenditure effect even when $B''(D) = 0$. A risk premium rising with indebtedness is sufficient to generate the concavity required for the threshold result, even when the benchmark funding rate is constant or falling.

The restriction $\rho'(D) > 0$ warrants comment, since much of the post-Keynesian literature treats the loan rate as set by lenders and has credit supply adjust through quantity rationing rather than price (Lavoie, 1996); both Ryoo and Kim (2014) and Ryoo (2016) adopt this convention explicitly. Under strict quantity rationing the price channel is absent, but the concavity required for Assumption 2 does not depend on it. Any mechanism making the marginal servicing burden increasing in the debt stock—a widening premium, tightening loan-to-value ratios, shortening maturities, or rising insurance and origination costs— delivers $F_{DD} < 0$ once it is entered in place of $\lambda(\theta)\rho(D)D$ in Equation (11). The risk-premium specification is adopted here because it yields the result in closed form, not because the price channel is the only admissible route. What the argument requires is Kalecki's principle of increasing risk in some form (Kalecki, 1937; Mott, 2009), not a particular institutional mechanism by which lenders impose it.

Under the pecking-order specification of Setterfield and Kim (2016), servicing is met out of borrowers' saving rather than their consumption. In the limiting case where saving fully absorbs servicing, the propensity applying to the outflow is zero rather than c_b, so that Equation (12) is replaced by $\lambda_S(\theta) = -c_c(\theta) < 0$ and the expression for F_D in Equation (14) becomes

$$F_D(D, \theta) = B'(D) + |\lambda_S(\theta)| \left[r_f + \rho(D) + D\rho'(D)\right] > 0 \qquad \text{for all } D. \tag{20}$$

Assumption 4 then fails: the marginal expenditure effect of debt is positive at every debt level, no interior threshold exists, and the specification reproduces the unambiguously expansionary debt effect that Setterfield and Kim (2016) obtain. The threshold result is therefore conditional on the disposition of debt service rather than on its existence. The two specifications are complementary: Setterfield and Kim (2016) characterize the growth regime that obtains when servicing is met from saving, whereas the threshold derived here characterizes the demand block when it is met from consumption. Which disposition prevails is an empirical question, and one that may itself depend on indebtedness: households that exhaust liquid saving may increasingly absorb servicing through reduced consumption, generating a transition toward the baseline regime at high indebtedness that would reinforce the concavity assumed in Assumption 2.

3. The Expansionary–Contractionary Debt Threshold

Differentiating Equation (6) with respect to household debt gives

$$\frac{\partial Y}{\partial D} = \mu(\theta) F_D(D, \theta). \tag{21}$$

Since $\mu(\theta) > 0$, the sign of the marginal demand effect of debt is determined by $F_D(D, \theta)$.

Theorem 1 (Unique debt threshold)**.** *Under Assumptions 1, 2, and 4, for every admissible $\theta \in (0, 1)$ there exists a unique debt level $\bar{D}(\theta) > 0$ satisfying*

$$F_D\big(\bar{D}(\theta), \theta\big) = 0. \tag{22}$$

Moreover,

$$\frac{\partial Y}{\partial D} \begin{cases} > 0, & D < \bar{D}(\theta), \\ = 0, & D = \bar{D}(\theta), \\ < 0, & D > \bar{D}(\theta). \end{cases} \tag{23}$$

Hence additional household borrowing is expansionary below the threshold and contractionary above it.

Proof. Assumption 1 states $F_D(0, \theta) > 0$, while Assumption 4 implies $F_D(D, \theta) < 0$ for sufficiently large D. Continuity of F_D therefore guarantees existence of a zero by the Intermediate Value Theorem. Assumption 2 implies $F_{DD} < 0$, so F_D is strictly decreasing in D and the zero is unique and strictly positive. Since $\mu(\theta) > 0$, Equation (21) gives the sign pattern in Equation (23). $\square$

Note that $\bar{D}(\theta)$ is determined entirely by F_D: the multiplier $\mu(\theta)$ scales the magnitude of $\partial Y/\partial D$ but not its sign, and therefore does not enter Equation (22). The distributional comparative static in Corollary 1 accordingly operates through the leakage term $\lambda(\theta)$ alone. Both channels nonetheless originate in the single primitive $c_c'(\theta) < 0$, which simultaneously lowers the multiplier and raises the fraction of debt-service income withdrawn from expenditure.

Corollary 1 (Income distribution)**.** *Under Assumption 3, the threshold satisfies*

$$\frac{d\bar{D}(\theta)}{d\theta} = -\frac{F_{D\theta}\big(\bar{D}(\theta), \theta\big)}{F_{DD}\big(\bar{D}(\theta), \theta\big)} < 0. \tag{24}$$

A larger creditor income share therefore causes the expansionary–contractionary transition to occur at a lower level of household debt.

Proof. Apply the Implicit Function Theorem to Equation (22). Since $F_{DD} < 0$ and $F_{D\theta} < 0$, the derivative in Equation (24) is negative. $\square$

The result concerns the location of a threshold in debt-stock space and is distinct from dynamic stability conditions in borrower–creditor cycle models.[5]

Corollary 2 (Critical collateral price)**.** *Under the binding collateral constraint in Equation (5), define*

$$\bar{q}(\theta) = \frac{\bar{D}(\theta)}{mH}. \tag{25}$$

For fixed m and H,

$$\frac{\partial Y}{\partial q} = \mu(\theta) F_D(mqH, \theta) mH, \tag{26}$$

[5] Palley (1994), for example, studies how distribution affects the stability region of a dynamic debt system. Corollary 1 instead concerns the debt level at which the marginal demand effect of additional borrowing changes sign.

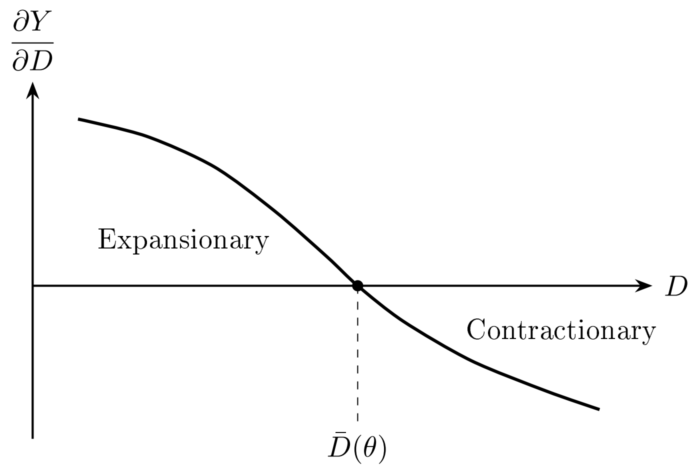

Figure 1: Expansionary–contractionary debt threshold. The marginal demand effect of household debt is positive below $\bar{D}(\theta)$ and negative above it. The diagram is schematic and does not impose a particular functional form.

and therefore

$$\frac{\partial Y}{\partial q} \begin{cases} > 0, & q < \bar{q}(\theta), \\ = 0, & q = \bar{q}(\theta), \\ < 0, & q > \bar{q}(\theta). \end{cases} \tag{27}$$

Moreover,

$$\frac{d\bar{q}(\theta)}{d\theta} = \frac{1}{mH}\frac{d\bar{D}(\theta)}{d\theta} < 0. \tag{28}$$

Thus, collateral appreciation raises aggregate demand below the critical collateral price but reduces it once q exceeds $\bar{q}(\theta)$.

Proof. From Equation (5), $\partial D/\partial q = mH > 0$. Applying the chain rule to Equation (6) yields Equation (26). The sign result follows from Theorem 1, while Equation (28) follows from Corollary 1. □

The collateral-price result isolates the borrowing-constraint channel. With H fixed, the model abstracts from other expenditure effects of collateral-price changes, including capital-gain redistribution, expectations, and refinancing, and allows q to affect demand through debt capacity in Equation (5).

Figure 1 illustrates the unique sign change established in Theorem 1.

4. Debt Dynamics under Collateral Feedback

The threshold in Theorem 1 is a property of the demand block. The following collateral-price closure preserves this threshold and gives it a simple dynamic interpretation.

Let the collateral price respond to demand pressure relative to a positive normal level of output $\bar{Y} > 0$:

$$\dot{q} = \kappa \left(Y - \bar{Y}\right), \qquad \kappa > 0. \tag{29}$$

With m and H fixed, the binding constraint in Equation (5) gives $\dot{D} = mH\dot{q}$, so that substituting Equation (6) yields a one-dimensional law of motion for household debt:

$$\dot{D} = mH\kappa \left[\mu(\theta)\left(A + F(D,\theta)\right) - \bar{Y}\right] \equiv \Phi(D,\theta). \tag{30}$$

Equation (30) closes the collateral–credit loop: borrowing raises demand, demand raises the collateral price, and a higher collateral price relaxes the borrowing constraint.

Differentiating Equation (30) with respect to D gives

$$\Phi_D(D,\theta) = mH\kappa\,\mu(\theta)\,F_D(D,\theta), \qquad \Phi_{DD}(D,\theta) = mH\kappa\,\mu(\theta)\,F_{DD}(D,\theta) < 0, \tag{31}$$

where the sign of Φ_{DD} follows from Assumption 2. Since $mH\kappa\mu(\theta) > 0$, the sign of Φ_D coincides with that of F_D. Hence Φ is strictly concave in D and attains a unique maximum at $\bar{D}(\theta)$, by Theorem 1: debt accumulates fastest precisely at the threshold.

Assumption 5 (Admissible normal output)**.**

$$\mu(\theta)\left[A + F(0,\theta)\right] < \bar{Y} < \mu(\theta)\left[A + F(\bar{D}(\theta),\theta)\right]. \tag{32}$$

The lower inequality states that demand at zero household debt falls short of normal output, so that some borrowing is required to sustain it; the upper inequality states that normal output is attainable, so that the debt-financed expansion does not grow without bound.

Proposition 1 (Accelerating and decelerating credit booms)**.** *Under Assumptions 1–4 and 5, Equation (30) has exactly two stationary points D_1 and D_2 with*

$$0 < D_1 < \bar{D}(\theta) < D_2,$$

where D_1 is unstable and D_2 is locally asymptotically stable. Moreover, along any expanding trajectory ($\dot{D} > 0$),

$$\ddot{D} = \Phi_D(D,\theta)\,\dot{D} \begin{cases} > 0, & D < \bar{D}(\theta), \\ = 0, & D = \bar{D}(\theta), \\ < 0, & D > \bar{D}(\theta). \end{cases}$$

Proof. Setting $\Phi(D,\theta) = 0$ in Equation (30) and dividing by $mH\kappa\mu(\theta) > 0$ gives $F(D,\theta) = \bar{Y}/\mu(\theta) - A$. By Assumptions 1, 2 and 4, F is strictly concave in D, increasing on $[0, \bar{D}(\theta))$ and decreasing on $(\bar{D}(\theta), \infty)$; since Assumption 4 bounds F_D above by a strictly negative constant for large D, $F \to -\infty$ as $D \to \infty$. Assumption 5 places $\bar{Y}/\mu(\theta) - A$ strictly between $F(0,\theta)$ and $F(\bar{D}(\theta),\theta)$, so the Intermediate Value Theorem delivers exactly one root on each branch, with $0 < D_1 < \bar{D}(\theta) < D_2$. Since $F_D > 0$ on $[0, \bar{D}(\theta))$ and $F_D < 0$ beyond it, Equation (31) gives $\Phi_D(D_1,\theta) > 0$ and $\Phi_D(D_2,\theta) < 0$, so D_1 is unstable and D_2 locally asymptotically stable. Finally, differentiating Equation (30) with respect to time gives $\ddot{D} = \Phi_D\dot{D}$, and the sign pattern follows from Theorem 1. □

The threshold therefore separates two phases of a single expansion rather than expansion from contraction. For $D \in (D_1, \bar{D}(\theta))$ the collateral–credit loop is self-reinforcing and debt accumulates at an increasing rate. For $D \in (\bar{D}(\theta), D_2)$ debt continues to accumulate but at a decreasing rate, and the economy converges to D_2 from below. For $D > D_2$ demand falls short of $\bar{Y}$, and debt declines toward D_2 from above. Below D_1 demand is insufficient to support the collateral price, and debt declines toward the lower boundary of the admissible debt domain. In this sense $\bar{D}(\theta)$ is the point at which a collateral-financed credit boom ceases to accelerate—a turning point derived from the expenditure function rather than imposed through expectations. Figure 2 illustrates this phase structure. The threshold appears there as the maximum of Φ rather than as the zero of $\partial Y/\partial D$ in Figure 1, since Φ_D is proportional to F_D by Equation (31).

Proposition 2 (Distribution and the accelerating phase)**.** *Under the specification in Equation* (11)*, $F_\theta = -\lambda'(\theta)r(D)D < 0$ for $D > 0$. Together with $\mu'(\theta) < 0$, this implies*

$$\frac{dD_1}{d\theta} > 0, \qquad \frac{d\bar{D}(\theta)}{d\theta} < 0, \qquad \frac{dD_2}{d\theta} < 0.$$

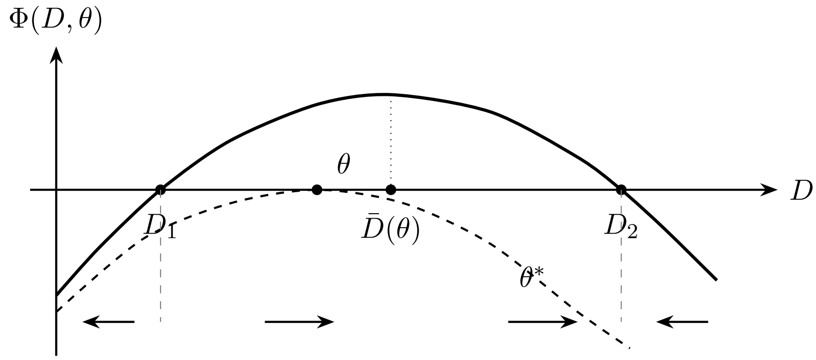

Figure 2: Debt accumulation under the collateral-price closure. The solid curve is $\Phi(D,\theta)$ for a creditor income share satisfying Assumption 5: debt accumulates on (D_1, D_2) and declines outside it, with $\bar{D}(\theta)$ the level at which accumulation is fastest. Arrows indicate the direction of motion along the solid curve. The dashed curve shows $\Phi(D,\theta^*)$, tangent to the axis at $\bar{D}(\theta^*) < \bar{D}(\theta)$; for $\theta > \theta^*$ no stationary point exists and debt declines from any initial position. Both diagrams are schematic and do not impose a particular functional form.

A larger creditor income share therefore contracts the accelerating interval $(D_1, \bar{D}(\theta))$ from both ends and lowers the stable debt level.

Proof. Write $G(D,\theta) \equiv \mu(\theta)[A + F(D,\theta)] - \bar{Y}$, so that stationary points solve $G = 0$. Then $G_D = \mu(\theta)F_D$ and $G_\theta = \mu'(\theta)[A+F] + \mu(\theta)F_\theta$. At any stationary point $A + F = \bar{Y}/\mu(\theta)$, hence $G_\theta = \mu'(\theta)\bar{Y}/\mu(\theta) + \mu(\theta)F_\theta < 0$. Applying the Implicit Function Theorem, $dD_i/d\theta = -G_\theta/G_D$, which is positive at D_1 (where $G_D > 0$) and negative at D_2 (where $G_D < 0$). The sign of $d\bar{D}/d\theta$ is Corollary 1. $\square$

Proposition 2 is local: it holds for variations in θ that preserve Assumption 5. Let $M(\theta) \equiv \mu(\theta)\left[A + F(\bar{D}(\theta),\theta)\right]$ denote the maximum demand attainable through the collateral–credit mechanism. Since $F_D(\bar{D}(\theta),\theta) = 0$, the envelope theorem gives $M'(\theta) = \mu'(\theta)\left[A+F\right] + \mu(\theta)F_\theta < 0$, so the upper bound in Equation (32) tightens as the creditor share rises. Starting from any θ satisfying Assumption 5, if $M(\theta)$ eventually falls below $\bar{Y}$, that is, if $\lim_{\theta\uparrow 1} M(\theta) < \bar{Y}$, there exists a unique θ^* at which D_1 and D_2 coalesce and beyond which Equation (30) admits no stationary point. For $\theta > \theta^*$ no level of household debt generates demand sufficient to sustain $\bar{Y}$, and debt declines toward the lower boundary of the admissible domain. Sufficiently concentrated creditor income can therefore render a collateral-financed expansion unsustainable, a conclusion reached by Setterfield and Kim (2016) through the debt-service feasibility ceiling rather than through the demand block.

Three qualifications are in order. First, the closure in Equation (29) treats $\bar{Y}$ as exogenous and omits expectations of capital gains; the resulting dynamics are monotone, and the model generates turning points but not endogenous cycles. Adding adaptive expectations of collateral appreciation, as in Ryoo (2016), would introduce a second state variable and, for sufficiently rapid expectation revision, a Hopf bifurcation and closed orbits. Second, the loan-to-value ratio m is held fixed; endogenous lending standards responding to the state of the balance sheet would constitute a further source of feedback. Neither of these two extensions alters the static threshold provided that it leaves the demand block $F(D, \theta)$ unchanged. Third, θ is treated as a parameter while D evolves. Since debt-service payments constitute income to creditors, endogenizing θ would introduce an additional distributional feedback and, in general, a second state variable. The omission is conditionally conservative for the present results: if higher indebtedness raises the creditor income share, Assumption 3 implies that this feedback would lower $\bar{D}(\theta)$ and, by Proposition 2, further compress the accelerating phase. Modeling this joint evolution is left for future work.

5. Discussion and Conclusion

The familiar expansionary and contractionary effects of household debt are not themselves new. The result established here is a debt-level threshold: when the marginal expenditure effect declines strictly with indebtedness, initial expansion and eventual contraction imply a unique $\bar{D}(\theta)$ at which additional borrowing changes sign. The risk-premium specification shows that this curvature can arise from increasing marginal servicing costs even when the gross marginal expenditure benefit of borrowing is constant.

Income distribution affects both the strength of demand and the location of this transition. A larger creditor income share lowers the aggregate multiplier and, when additional creditor income accrues to progressively lower-spending households, increases the fraction of debt-service income not recycled into expenditure. The debt threshold and corresponding critical collateral price therefore fall with θ, holding lending standards and the collateral stock fixed.

These effects extend to the dynamics of debt accumulation. Under the collateral-price closure of Section 4, $\bar{D}(\theta)$ is the debt level at which accumulation is fastest, and by Proposition 2 a distributional shift toward creditors contracts from both ends the interval of debt levels over which a collateral-financed expansion remains self-reinforcing, while lowering the stable debt level. An economy in which the creditor income share has risen therefore exhausts the accelerating phase of a credit boom at a lower level of household indebtedness. Distributional change and the durability of credit-driven growth are in this sense two aspects of the same mechanism, a connection reached here through the demand block rather than through the debt-servicing behaviour emphasized by Setterfield and Kim (2016).

Through the borrowing-constraint channel modeled here, collateral appreciation is not unconditionally expansionary. Higher collateral values raise demand below $\bar{q}(\theta)$ but reduce it above that threshold. Since $\bar{q}(\theta) = \bar{D}(\theta)/(mH)$, a higher loan-to-value ratio lowers the collateral price at which a given debt threshold is reached.

The contrast with Ryoo (2016) is instructive. There, the effect of housing wealth on aggregate consumption, $\chi_{h^w} = \mu_\omega$, is positive at every level of housing wealth, because the collateral channel enters credit supply linearly and the resulting boom is terminated by expectational dynamics rather than by any property of the collateral effect itself. Here the reversal is internal to the borrowing-constraint channel: concavity of F in D, transmitted through Equation (5), makes the sign of $\partial Y/\partial q$ depend on the level of q. The model correspondingly abstracts from the other asset-price channels that operate in Ryoo (2016) and elsewhere, including capital-gain redistribution, expectations of capital gains, and refinancing; each would add a distinct source of turning points.

The threshold $\bar{D}(\theta)$ should not be interpreted as a universal numerical debt limit. Its location depends on income distribution, the servicing schedule, and the expenditure benefit of credit. Empirical debt-to-GDP thresholds therefore motivate the nonlinear mechanism rather than directly identify the theoretical threshold. An empirical extension could estimate how such thresholds vary with distributional and financial conditions across economies.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Appendix A. Derivation of the Aggregate-Demand Representation

For completeness, the aggregate-demand representation in Equation (6) can be obtained from a conventional Keynesian expenditure system. Let autonomous expenditure excluding the first-round effect of household debt be denoted by A. Borrowers and creditors have marginal propensities to consume c_b and $c_c(\theta)$, respectively, and creditors receive the income share θ. Aggregate induced consumption is therefore

$$C^I = [(1-\theta)c_b + \theta c_c(\theta)]\, Y = c(\theta)Y, \tag{A.1}$$

where

$$c(\theta) = (1-\theta)c_b + \theta c_c(\theta). \tag{A.2}$$

Let $F(D, \theta)$ denote the net first-round expenditure effect associated with household indebtedness. Goods-market equilibrium then requires

$$Y = c(\theta)Y + A + F(D, \theta). \tag{A.3}$$

Rearranging gives

$$Y = \frac{1}{1-c(\theta)}\,[A + F(D,\theta)] = \mu(\theta)\,[A + F(D,\theta)]\,, \tag{A.4}$$

where

$$\mu(\theta) \equiv \frac{1}{1-c(\theta)} > 0.$$

Thus, the multiplier structure in Equation (6) is conventional. The substantive restriction introduced in the analysis concerns the dependence of the net expenditure term $F(D, \theta)$ on the level of household indebtedness.

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